HDOC- 6 – 2
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Program to Check Whether a Number is Positive, Negative or Zero Using Nested if–else

This document provides a clean, well-commented implementation for checking whether an integer is positive, negative or zero in C. The program takes an integer as input and uses a nested if–else structure to decide and display the result.

1. C Implementation

Step 1: Understand the Logic

- A number greater than 0 is positive and a number less than 0 is negative.
- The number 0 is neither positive nor negative.
- The outer if checks whether the number is greater than or equal to 0; the inner if–else then separates zero from positive.

Step 2: Define Variables and Data Types

- Use an integer variable (int) to store the input number.

Step 3: Take User Input

- Display a prompt using printf("Enter an integer: "), then read the value using scanf("%d", &num).

Step 4: Apply the Nested if–else Condition

1. Outer if: if num >= 0, go inside and check further; otherwise the number is negative.
2. Inner if: if num == 0, the number is zero.
3. Inner else: the number is positive.

Step 5: Format the Output

- Print the number along with the message "zero", "positive number" or "negative number" using printf().

Complete C Code

```c
/*
 * Q7 (User Inputs, Operations & Output)
 * Write a program to input an integer and check whether it is
 * positive, negative or zero using nested if-else.
 */
#include <stdio.h>

int main() {
    int num;

    // Step 1: Read the integer
    printf("Enter an integer: ");
    scanf("%d", &num);

    // Step 2: Check the sign using nested if-else
    if (num >= 0) {
        if (num == 0) {
            printf("%d is zero.\n", num);
        } else {
            printf("%d is a positive number.\n", num);
        }
    } else {
        printf("%d is a negative number.\n", num);
    }

    return 0;
}
```

Program Output (Sample Runs)

Run 1:

```
$ gcc hdoc6_2.c -o hdoc6_2
$ ./hdoc6_2
Enter an integer: 25
25 is a positive number.
```

Run 2:

```
$ ./hdoc6_2
Enter an integer: -8
-8 is a negative number.
```

Run 3:

```
$ ./hdoc6_2
Enter an integer: 0
0 is zero.
```

Dry Run with Test Cases

Sample Test Case 1:

- Input: 25
- Calculation:
  - Outer if: 25 >= 0 is TRUE, so go inside
  - Inner if: 25 == 0 is FALSE, so the inner else runs
- Output: 25 is a positive number.

Sample Test Case 2:

- Input: -8
- Calculation:
  - Outer if: -8 >= 0 is FALSE, so the outer else runs
- Output: -8 is a negative number.

Sample Test Case 3:

- Input: 0
- Calculation:
  - Outer if: 0 >= 0 is TRUE, so go inside
  - Inner if: 0 == 0 is TRUE, so the inner if block runs
- Output: 0 is zero.